Statistical Signal Processing
Lecture 07:
Unscented ("sigma-point") Kalman filter for non-linear systems

Literature and Problems: See course schedule

Agenda (preliminary):

Non-linear process model: $\mathbf{x_k} = \mathbf{f}(\mathbf{x_{k-1}}, \mathbf{u_{k-1}}) + \mathbf{w_{k-1}}$
Non-linear obs model: $\mathbf{z_k} = \mathbf{h}(\mathbf{x_k}) + \mathbf{v_k}$

EKF approach: linearize around the currently predicted state

- **only correct up to first order in Taylor series expansion for non-linearities**
- **requires analytical or numerical calculation of Jacobians** $\mathbf{A_k}, \mathbf{W_k}, \mathbf{H_k}, \mathbf{V_k}$
- **can diverge / not precise enough for even moderate non-linearity**

General problem formulation given non-linearity $\mathbf{y} = \mathbf{f}(\mathbf{x}), \quad \mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ **:**
Propagate mean, covariance (and pdf) for random vector X through non-linearity

- **in general not possible analytically (except special cases)**
- **EKF approach:** $\bar{\mathbf{y}} = \mathbf{f}(\bar{\mathbf{x}})$, **BUT: SLIDE*2: Jensen's inequality + simple "proof"**
- **Taylor series for** $\mathbf{y} = \mathbf{f}(\bar{\mathbf{x}} + \Delta\mathbf{x})$ **giving** $\bar{\mathbf{y}} = \mathbf{f}(\bar{\mathbf{x}})$ **and** $\mathbf{Cov}[\mathbf{y}] = \nabla\mathbf{f}\,\mathbf{Cov}[\mathbf{x}]\,\nabla\mathbf{f}^{\mathbf{T}}$
- **refer to EKF estimates** $\hat{\mathbf{x}}_\mathbf{k}^- = \mathbf{f}(\hat{\mathbf{x}}_{\mathbf{k-1}})$ **and** $\mathbf{P}_\mathbf{k}^- = \mathbf{A_k}\mathbf{P_{k-1}}\mathbf{A_k}^{\mathbf{T}}$
- **DEMO: 2D example of propagation of moments through non-linearity**

Unscented Transform (UT), ("Sigma-point transform"): alternative approach

- **basic idea: define x-sigma-points around** $\bar{\mathbf{x}}$**, propagate these through non-linearity to y-sigma-points, estimate** $\bar{\mathbf{y}}$ **and** $\mathbf{P_y} = \mathbf{Cov}[\mathbf{y}]$ **from y-sigma-points**
- **non-linearity is not approximated and no Jacobians required**
- **UT in more detail: Given** $\bar{\mathbf{x}}, \mathbf{P_x}$**:** $\mathbf{x}^{(\mathbf{i})} = \bar{\mathbf{x}} + \tilde{\mathbf{x}}^{(\mathbf{i})} \rightarrow \mathbf{y}^{(\mathbf{i})} = \mathbf{f}(\mathbf{x}^{(\mathbf{i})}) \xrightarrow{\mathbf{w}^{(\mathbf{i})}} \bar{\mathbf{y}}_\mathbf{u}, \mathbf{P}_{\mathbf{y,u}}$
- **SLIDE*3: Visualization of UT approach**
- **technicalities about matrix square root**
- **x-sigma-points are chosen so that** $\mathbf{E}[\mathbf{x}^{(\mathbf{i})}] = \bar{\mathbf{x}}$ **and** $\mathbf{Cov}[\mathbf{x}^{(\mathbf{i})}] = \mathbf{P_x}$ **(proof?)**
- **UT performance:** $(\bar{\mathbf{y}}_\mathbf{u}, \mathbf{P}_{\mathbf{y,u}})$ **estimates** $(\mathbf{y}, \mathbf{P_y})$ **up to 3'rd order (EKF only 1'st)**
- **DEMO again: Comparison of UT and EKF approaches for 2D example**

At last: UT used in Kalman filter algorithm → UKF, Unscented Kalman filter

- **SLIDE: UKF algorithm**
- **UKF vs. EKF advantages**
- **non-additive noise case**
- **generalized version of UT/UKF (can give marginal improvement)**

DEMO: UKF used for state estimation for falling body (3D example)
DEMO: UKF used for state estimation for water tank system (1D example)